

variable but steady, like pension costs; or contingent one-off payments, like a person's death benefit.

When liabilities are introduced, the optimal portfolio strategy has three components:

$$\begin{aligned} \text{Long-Run Weight (t)} = & \text{Liability Hedge (t)} \\ & + \underbrace{\text{Short-Run Weight (t)} + \text{Opportunistic Weight (t)}}_{\text{Investment Portfolio (t)}}. \end{aligned} \quad (4.23)$$

The investment portfolio is exactly the same as the nonliability case that we examined in sections 2 and 3: the optimal policy is to rebalance under i.i.d. returns and when returns are predictable, the optimal short-run portfolio changes over time, and the long-run investor has additional opportunistic strategies. The liability hedging portfolio is the portfolio that best ensures the investor can meet those liabilities. We solve for it by holding asset positions that produce the highest correlation with the liabilities.³³

There are several special cases of optimal liability hedging portfolios:

1. Cash flow matching or immunization.

This involves constructing a perfect match of liability outflows each period. You immunize each liability cash flow by holding bonds of appropriate maturities.

2. Duration matching.

If liabilities can be summarized by a single interest rate factor, which is common for pension liabilities, then the liabilities can be offset by an asset portfolio with the same duration.³⁴

3. Liability-driven investing.

This aims to construct a portfolio of risky assets that best meets the liability obligations. It is also common in pension fund management and was introduced by Sharpe (1992). It is related to and often used synonymously with . . .

4. Asset–liability matching.

This is a more general case than duration matching. In asset–liability matching, dimensions other than just duration are used to match liability characteristics with assets, including liquidity, sensitivity to factors besides only interest rates, and horizon.

The Merton–Samuelson advice of long-horizon asset allocation extended to liabilities is, first, to meet the liabilities and then to invest the excess wealth over the present value of liabilities in the same style as sections 2 and 3, using the myopic market portfolio and the opportunistic long-horizon portfolio.

³³ It was also introduced by the grandfather of portfolio choice theory, Merton, in 1993.

³⁴ Duration is exposure to the interest rate level factor, which is the most important factor in fixed income investments. See chapter 9.